

Two players take turns taking counters from a pile. On each turn, a player takes 1 counter or 2 counters. Whoever takes the last counter wins.

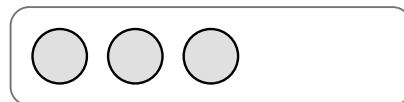
**Problem 1:** It is your turn. Circle the counters you take to be sure to win, or cross out the pile if you cannot be sure to win.



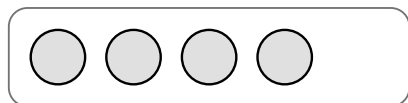
1



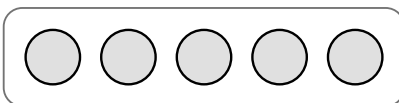
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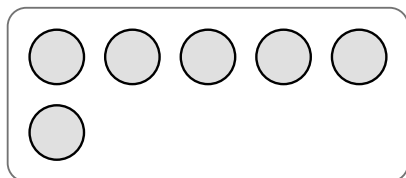


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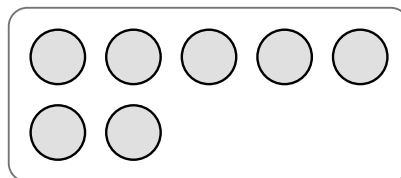
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**Problem 2:** Circle 1st or 2nd to show whether you would rather go first or second with each pile.



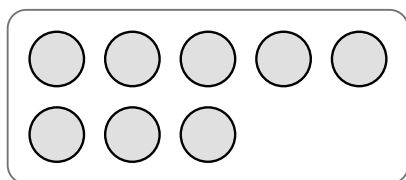
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1st      2nd



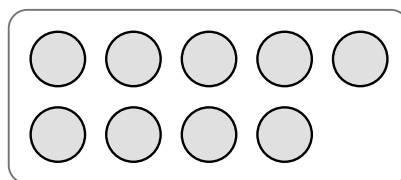
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1st      2nd



8

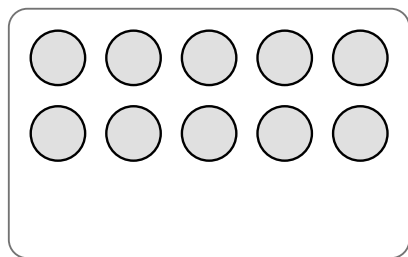
1st      2nd



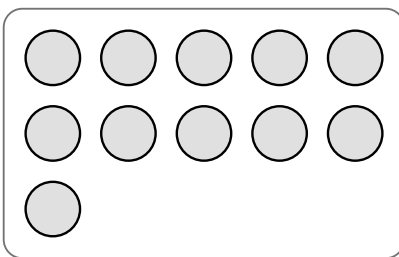
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1st      2nd

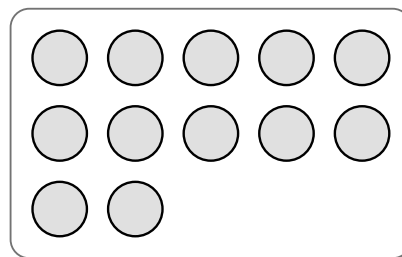
**Problem 3:** It is your turn with a bigger pile. Circle the counters you take to be sure to win, or cross out the pile if you cannot be sure to win.



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11



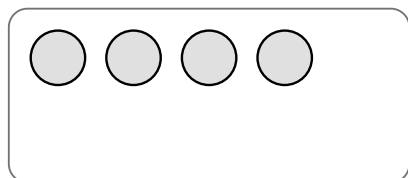
12

**Problem 4:** A pile is a trap if the player whose turn it is cannot be sure to win. Put an X on every trap from 1 to 20.

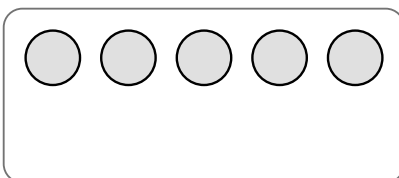
|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|----|

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|----|----|----|----|----|----|----|----|----|----|

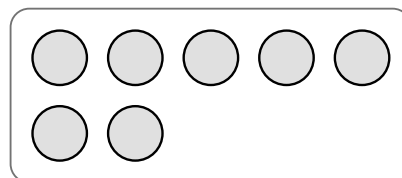
**Problem 5:** Ben always starts by taking 2 counters. Circle each pile where Ben can still be sure to win after that.



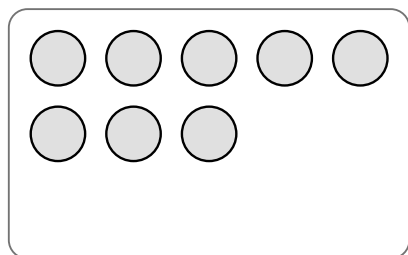
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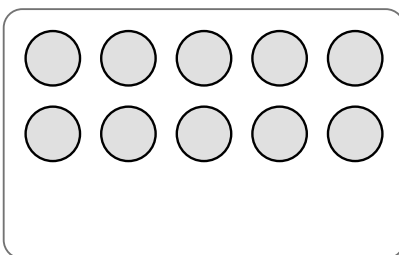
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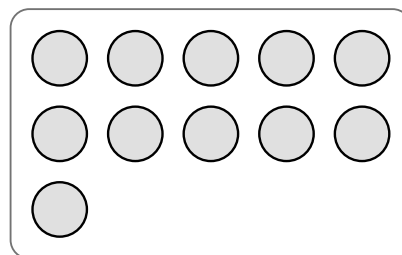
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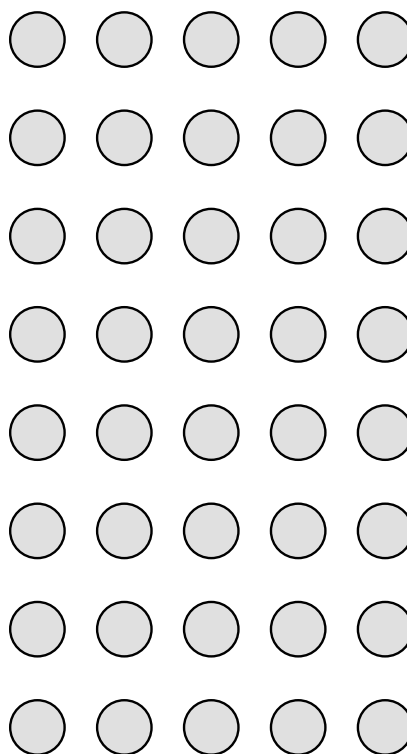
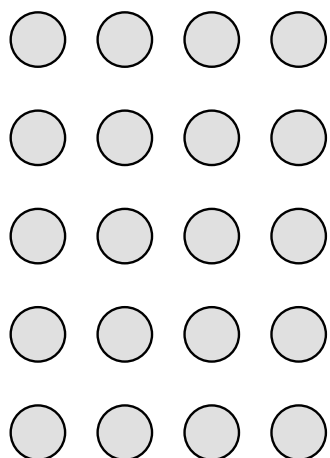


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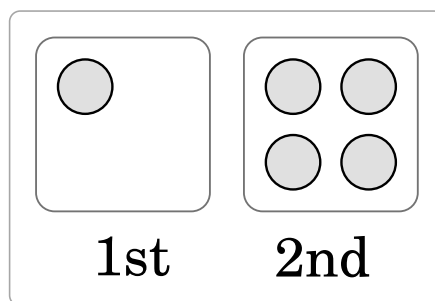
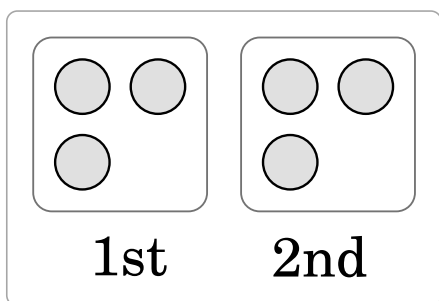
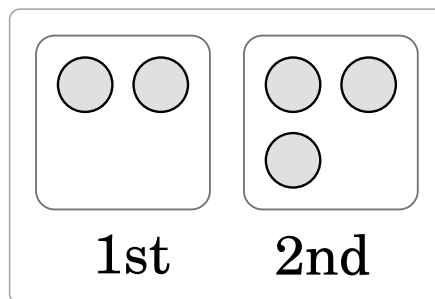
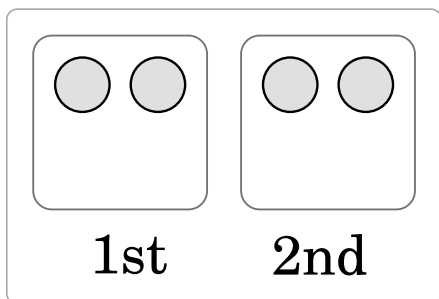
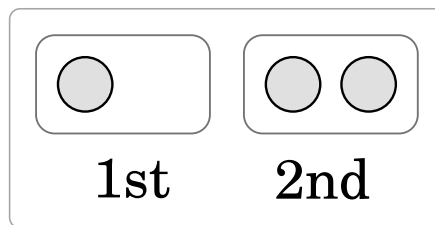
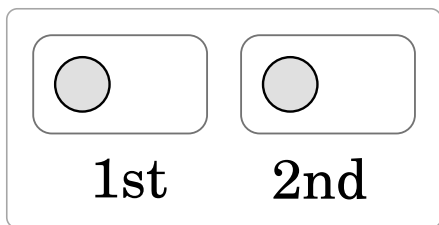
**Problem 6:** Show every different way a game with 4 counters can go, and then every way a game with 5 counters can go. Circle the counters taken on each turn.



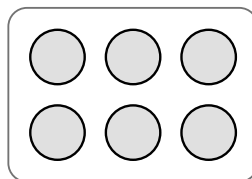
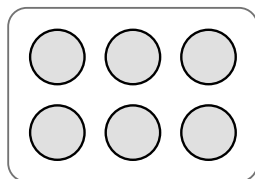
**Problem 7:** Now each player may take 1, 2, or 3 counters on a turn. Put an X on every trap from 1 to 20.

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |

**Problem 8:** Now there are two piles, and on your turn you take 1 or 2 counters from one pile. Whoever takes the last counter on the table wins. Circle 1st or 2nd to show whether you would rather go first or second.



**Problem 9:** Play the game from Problem 8 with two piles of 6 counters. If you go second, how can you win every time, whatever the other player does?



**Problem 10:** Now there is one pile again, and whoever takes the last counter loses. Each player may take 1 or 2 counters on a turn. Put an X on every trap from 1 to 20.

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |